

Example of inductive definition

Example

The set Nat of natural numbers is defined **inductively** as follows:

- ① $0 \in Nat$
- ② if $x \in Nat$ then $s(x) \in Nat$

Note that with the inductive definition of Nat , we can also

- define inductively any function F of the form $F : Nat \longrightarrow C$, using the following scheme:
 - ① **Base step** We directly provide the definition of $F(0)$
 - ② **Inductive step** Starting from $F(x)$, we provide the definition of $F(s(x))$
- prove that a certain property T holds for **every** natural numbers, using the following scheme:
 - ① **Base step** We directly prove that $T(0)$ holds
 - ② **Inductive step** Based on the assumption that $T(x)$ holds, we provide the proof that $T(s(x))$ holds as well

Syntax of propositional logic language: the formulas

Definition (Well formed formulas (or simply formulas))

A propositional logic formula on the alphabet $\mathcal{P}$ is a finite expression built according to the following **inductive** definition:

- every $P \in \mathcal{P}$ is an **atomic formula**, and every atomic formula is a **formula**
- if A and B are formulas, then the following are **formulas**:
 - ▶ (A)
 - ▶ $\neg A$
 - ▶ $A \wedge B$
 - ▶ $A \vee B$
 - ▶ $A \supset B$ (also written as $A \rightarrow B$)
 - ▶ $A \equiv B$

Example (formulas and non formulas)

Formulas	Non formulas
$P \rightarrow Q$	PQ
$P \rightarrow (Q \rightarrow R)$	$(P \rightarrow \wedge((Q \rightarrow R)$
$P \wedge Q \rightarrow R$	$P \wedge Q \rightarrow \neg R \neg$

Reading formulas

Problem

How do we read the formula $P \wedge Q \rightarrow R$?

The formula $P \wedge Q \rightarrow R$ can be read in two ways:

- ❶ $(P \wedge Q) \rightarrow R$
- ❷ $P \wedge (Q \rightarrow R)$

Symbol priority

$\neg$ has higher priority, then $\wedge$, $\vee$, $\rightarrow$ and $\equiv$. Parenthesis can be used around formulas to stress or change the priority.

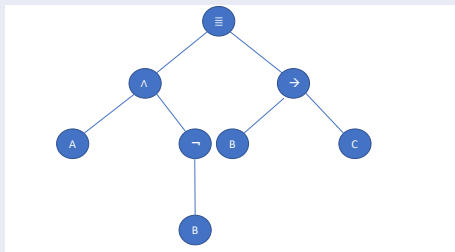
Symbol	Priority
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\equiv$	5

Formulas as trees

Tree form of a formula

A formula can be seen as a tree. Leaf nodes are associated to propositional variables, while intermediate (non-leaf) nodes are associated to connectives.

For instance the formula $(A \wedge \neg B) \equiv (B \rightarrow C)$ can be represented as the tree



Subformulas

Definition ((Proper) Subformula)

- A is a **subformula** of itself
- A, B are **subformulas** of $A \wedge B, A \vee B, A \supset B, A \rightarrow B, A \equiv B$
- A is a subformula of $\neg A$, and A is a subformula of (A)
- if A is a subformula of B and B is a subformula of C , then A is a subformula of C .
- A is a **proper subformula** of B if A is a subformula of B and A is different from B .

Remark

The subformulas of a formula represented as a tree correspond to all the different subtrees of the tree associated to the formula, one for each node.

Subformulas

Example

The subformulas of $(p \rightarrow (q \vee r)) \rightarrow (p \wedge \neg p)$ are

$$\begin{aligned} & (p \rightarrow (q \vee r)) \rightarrow (p \wedge \neg p) \\ & (p \rightarrow (q \vee r)), \quad (p \wedge \neg p), \quad p \rightarrow (q \vee r), \quad (q \vee r) \\ & p \wedge \neg p, \quad q \vee r \\ & p, \quad \neg p, \quad q, \quad r \end{aligned}$$

Proposition

Every formula has a finite number of subformulas

Note that the above is a **meta-sentence**, i.e., a sentence telling something about the formal logical system. In general we should prove that a proposition (a meta-sentence) is true. When we do so, we use logic for proving properties of logic!

Propositional logic language: semantics

How can we specify the meaning of the sentences of propositional logic?

- The simplicity of the propositional language leads naturally to deciding that the meaning of any sentence coincides with its truth value, i.e., is either true or false.
- But, for example, a sentence like $P_1 \wedge P_2$ is either true or false **depending** of the truth value of P_1 and P_2 . So, the meaning of a sentence depends on the truth value of the propositional variables in it!
- In turn, the truth value of a propositional variable depends on the specific intended meaning of the variable in the considered context. For example, if “piove” is a propositional variable, its truth value depends on the “world” (context, i.e., time and space) that we are considering.
- In order to capture the notion of world (or, context), propositional logic introduces the notion of **interpretation**.

Propositional logic language: semantics

The notion of interpretation is central to the semantics of propositional logic.

Definition (Propositional interpretation (or simply interpretation))

Given a propositional alphabet with propositional variables $\mathcal{P}$, a **propositional interpretation** is a function $\mathcal{I} : \mathcal{P} \rightarrow \{\text{True}, \text{False}\}$

Remark

A propositional interpretation can be actually based on the subset T of $\mathcal{P}$ containing exactly the propositional variables that are true in the context corresponding to the interpretation. From this point of view, the interpretation $\mathcal{I}$ can be thought of as the **characteristic function** of T , i.e., the function satisfying the condition $\mathcal{I}(P) = \text{True}$ if and only if $P \in T$.

Interpretation in propositional logic: example

Consider the following sentence over the alphabet $\{p, q, r\}$:

$$(p \wedge (\neg q \vee r)) \vee (p \wedge r)$$

Some possible interpretations for the above sentence:

Example

	p	q	r	Set theoretic representation
$\mathcal{I}_1$	True	True	True	$\{p, q, r\}$
$\mathcal{I}_2$	True	False	True	$\{p, r\}$
$\mathcal{I}_3$	True	False	False	$\{p\}$
$\mathcal{I}_4$	False	False	False	$\{\}$

In the above picture, the table should be read as follows: $\mathcal{I}_1$ is the interpretation function such that $\mathcal{I}_1(p) = \text{True}$, $\mathcal{I}_1(q) = \text{True}$, $\mathcal{I}_1(r) = \text{True}$, and so on.

Evaluation of a propositional formula in an interpretation

We now illustrate the rules for deciding whether the formula A is true in the interpretation $\mathcal{I}$ (or, equivalently, whether $\mathcal{I}$ satisfies A , written as $\mathcal{I} \models A$).

Definition ($\mathcal{I} \models A$)

$\mathcal{I} \models A$ is inductively defined as follows:

- If $P \in \mathcal{P}$, $\mathcal{I} \models P$ if $\mathcal{I}(P) = \text{True}$
- $\mathcal{I} \models (A)$ if $\mathcal{I} \models A$
- $\mathcal{I} \models \neg A$ if it is not the case that $\mathcal{I} \models A$ (also written $\mathcal{I} \not\models A$)
- $\mathcal{I} \models A \wedge B$ if both $\mathcal{I} \models A$ and $\mathcal{I} \models B$
- $\mathcal{I} \models A \vee B$ if $\mathcal{I} \models A$ or $\mathcal{I} \models B$
- $\mathcal{I} \models A \rightarrow B$ if it is not the case that $\mathcal{I} \models A$ and $\mathcal{I} \not\models B$
- $\mathcal{I} \models A \equiv B$ if both $\mathcal{I} \models A \rightarrow B$ and $\mathcal{I} \models B \rightarrow A$

Evaluation of a propositional formula: example

Example

Let $\mathcal{I} = \{P\}$, and let us check if $\mathcal{I} \models (P \wedge Q) \vee (R \rightarrow S)$.

We replace each occurrence of each primitive proposition of the formula with the truth value assigned by $\mathcal{I}$, and apply the definition for connectives.

$$(True \wedge False) \vee (False \rightarrow False) \quad (1)$$

$$False \vee True \quad (2)$$

$$True \quad (3)$$

Observations on connectives

- $A \rightarrow B$ has the same meaning as $\neg A \vee B$
- $\neg(A \vee B)$ has the same meaning as $\neg A \wedge \neg B$ (De Morgan law 1)
- $\neg(A \wedge B)$ has the same meaning as $\neg A \vee \neg B$ (De Morgan law 2)
- $A \rightarrow B$ has the same meaning as $\neg B \rightarrow \neg A$
- $A \rightarrow \neg B$ has the same meaning as $B \rightarrow \neg A$
- A is a sufficient condition for B can be written as $A \rightarrow B$, and can read as “if A then B ”
- A is a necessary condition for B can be written as $B \rightarrow A$
- A is a necessary and sufficient for B can be written as $A \equiv B$, and can be read as “ A if and only if B ”
- **NAND** is another connective that is defined as $A \text{ NAND } B$ is defined as $\neg(A \wedge B)$
- **NOR** is another connective that is defined as $A \text{ NOR } B$ is defined as $\neg(A \vee B)$

Exercise

Prove the following proposition (i.e., prove that the following meta-sentence is true).

Proposition

Let $\mathcal{I}$ and $\mathcal{I}'$ be two interpretations for a formula A . If for every propositional variable P appearing in A it holds that $\mathcal{I}(P) = \mathcal{I}'(P)$, then $\mathcal{I} \models A$ if and only if $\mathcal{I}' \models A$.

Algorithm for checking if $\mathcal{I} \models A$

Lazy evaluation algorithm (1/2)

$(A \text{ is } p)$	<pre>check($\mathcal{I} \models p$) if $\mathcal{I}(p) = \text{true}$ then return YES else return NO</pre>
$(A \text{ is } B \wedge C)$	<pre>check($\mathcal{I} \models B \wedge C$) if check($\mathcal{I} \models B$) then return check($\mathcal{I} \models C$) else return NO</pre>
$(A \text{ is } B \vee C)$	<pre>check($\mathcal{I} \models B \vee C$) if check($\mathcal{I} \models B$) then return YES else return check($\mathcal{I} \models C$)</pre>